# Machine learning for prognostic prediction in heart and lung transplantation: a systematic review and meta-analysis of the donor–recipient dual-risk problem

Full study protocol

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PROSPERO registration draft initiated: 22 October 2025
Current structured protocol: Version 2.1, 13 October 2026 (erratum to Version 1.0 of 11 August 2026)

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# 1. Administrative information

## 1.1 Title

Machine learning for prognostic prediction in heart and lung transplantation: a systematic review and meta-analysis of the donor–recipient dual-risk problem.

## 1.2 Registration

PROSPERO. Registration identifier: **CRD420251174511.**

## 1.3 Authors and affiliations


| Role | Name | ORCID | Affiliation |
|---|---|---|---|
| Guarantor, named contact | Professor Johan Nilsson | 0000-0001-6860-6090 | Lund University, Faculty of Medicine, Department of Translational Medicine, Thoracic Surgery and Bioinformatics Research Unit, Lund, Sweden |
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| Protocol author; abstract screening | Eliot Peyster | 0000-0002-3350-3136 | University of Pennsylvania, Philadelphia, Pennsylvania, USA; Brisbane, Australia |

PROSPERO review team at initial registration: Johan Nilsson and Francesca Lunardi. Additional protocol authors contributed to abstract screening and may be added to the PROSPERO record in a subsequent amendment.

Correspondence: johan.nilsson@med.lu.se

Countries represented by the PROSPERO review team at initial registration: Sweden and Italy.

Search strategy design and execution: Matthias Bank, librarian and information specialist, Library & ICT, Faculty of Medicine, Lund University. He designed the search strategy, ran the initial search in full.

## 1.4 Protocol history

An initial review protocol and review framework were established by 14 June 2024, when the first literature search had been completed and the review was registered in Covidence. The review scope, eligibility criteria, and core search concepts were defined at that stage, and the screening workflow was documented contemporaneously in Covidence. A PROSPERO registration draft was initiated and timestamped on 22 October 2025. The protocol was subsequently refined and formalised into the present structured version without changing the fundamental review question or eligibility framework.

| Date | Protocol/update | Description |
|---|---|---|
| 14 June 2024 | Initial protocol | Initial review protocol/framework established; review scope, eligibility criteria, and core search strategy defined. |
| 14 June 2024 | Initial search and screening | Initial literature search completed and review registered in Covidence; screening workflow documented prospectively. |
| 1–6 August 2025 | Search update | Literature search updated using the same core search concepts and eligibility framework. |
| 22 October 2025 | PROSPERO draft initiated | PROSPERO registration draft created and timestamped; not submitted for publication. |
| 24 July 2026 | Search update | Literature search updated using the same core search concepts and eligibility framework. |
| 11 August 2026 | Structured protocol v1.0 | Protocol formalised and structured for PROSPERO registration and public archiving. |
| 27 September 2026 | Version 2.0 (erratum) | §8: reference-list checking removed (entered in error; never part of the |

| | | |
|---|---|---|
| | | review). Administrative updates: PROSPERO identifier inserted (CRD420251174511); ORCIDs added for Zaffiri and Peyster. No change to methods. |
| 13 October 2026 | Version 2.1 (erratum) | §7: the remaining mention of reference-list searching removed, consistent with Version 2.0. ORCIDs added for Hsich. No change to methods. |

### 1.5 Amendments

Any subsequent amendment to this structured protocol will be recorded in the PROSPERO record with its date and rationale. The protocol will be made available in full after the review is completed.

### 1.6 Support

The review has no specific external funding and is supported by the guarantor's and review team's non-commercial institutions. No funder has any role in the design, conduct, analysis or reporting of the review.

## 2. Background

Heart and lung transplantation require complex clinical decisions involving donor evaluation, candidate selection, donor–recipient matching, waitlist management, and prediction of post-transplant outcomes. Machine-learning methods are increasingly applied to these tasks because they can model complex, non-linear relationships that may not be captured by conventional statistical models.

Allocation in thoracic transplantation depends on a joint judgement about two people. A donor organ of a given quality performs differently in different recipients, and a recipient of a given risk fares differently with different organs. This dual dependence is the feature that additive linear risk scores handle least well, and it is the specific reason machine-learning approaches have been proposed in this field.

## 3. Rationale

Interest in artificial intelligence and machine learning in thoracic transplantation has increased substantially in recent years. It remains unclear how much of this literature represents original, peer-reviewed clinical research rather than abstracts, reviews or exploratory work, and whether published models have demonstrated clinically meaningful benefit. Preliminary screening indicates that broad searches identify a large number of records, while only a small proportion meet strict eligibility criteria for original clinical studies.

Important uncertainties remain regarding study quality, risk of bias, prospective evaluation, external validation, clinical implementation, and whether sufficient homogeneous evidence exists to permit quantitative synthesis. This review will evaluate the quantity, quality, methodological rigour, predictive performance, validation and clinical maturity of AI/ML research in thoracic transplantation and identify evidence gaps and priorities for clinical translation.

# 4. Objectives

The primary objective is to evaluate the current evidence on the use of artificial intelligence and machine learning in adult thoracic transplantation (heart, lung, and heart–lung transplantation).

The specific objectives are to:

1. systematically identify and characterise original studies evaluating AI/ML models across the thoracic transplantation pathway, including donor evaluation, candidate selection, waitlist management, donor–recipient matching, and prediction of post-transplant outcomes;
2. assess the methodological quality, risk of bias, validation strategies and clinical maturity of published AI/ML models;
3. quantitatively synthesise the predictive performance of comparable AI/ML models by meta-analysis where sufficient homogeneous data are available;
4. identify evidence gaps and priorities for future research and clinical implementation.

# 5. Review question

What is the role of artificial intelligence and machine learning in thoracic transplantation — including donor evaluation, candidate selection, waitlist management, donor–recipient matching and post-transplant outcomes — compared with current clinical practice?

# 6. Condition or domain being studied

This review concerns the application of artificial intelligence and machine learning in adult thoracic transplantation, including heart, lung and heart–lung transplantation. It encompasses the entire transplant pathway: donor evaluation, transplant candidate selection, waitlist management, donor–recipient matching, and prediction of post-transplant outcomes including survival, complications, graft rejection, quality of life and health-economic outcomes. Only studies in which a non-linear machine-learning model constitutes the primary analytical component are eligible.

# 7. Eligibility criteria

## 7.1 Population

**Included.** Adults (≥18 years) who are candidates listed for heart, lung or heart–lung transplantation; recipients of heart, lung or heart–lung transplantation; or heart and/or lung donors. Studies including both adults and paediatric participants are included only if data for adults can be extracted separately.

**Excluded.** Animal studies; paediatric-only populations (<18 years); studies of non-thoracic transplantation; and studies evaluating patients receiving alternative therapies instead of transplantation, such as a left ventricular assist device as destination therapy.

## 7.2 Intervention or exposure

**Included.** Studies evaluating artificial intelligence or machine learning as the primary analytical component. Eligible models are non-linear computational models, including but not limited to decision trees, random forests, gradient boosting, neural networks, deep learning, support vector machines and other non-linear machine-learning algorithms applied to prediction or decision support in thoracic transplantation.

**Excluded.** Studies evaluating only conventional linear statistical models, such as logistic regression or Cox proportional-hazards regression, without a machine-learning component. Studies in which machine learning is used only as a secondary analytical tool — for example to analyse omics, imaging or other data generated by a primary non-ML method — rather than as the primary analytical component.

## 7.3 Comparator

**Included.** Where available, comparators include current clinical practice, conventional statistical prediction models such as logistic or Cox regression, established clinical risk scores, clinician assessment, or alternative machine-learning models. **Studies without an explicit comparator are eligible** if they develop or validate an eligible machine-learning prediction model.

**Excluded.** None. Studies are not excluded based on comparator. Both comparative and non-comparative studies are eligible provided all other criteria are met.

## 7.4 Outcomes

At least one of: post-transplant survival; waitlist mortality; post-transplant complications; quality of life; health-economic outcomes.

## 7.5 Study design

Both randomised and non-randomised designs are eligible: randomised controlled trials; prospective and retrospective observational studies; registry-based studies; cohort studies; and other original peer-reviewed studies that develop, validate or evaluate a non-linear machine-learning model in adult thoracic transplantation.

**Excluded.** Conference abstracts and proceedings; conference papers; case reports; narrative reviews; systematic reviews; meta-analyses; editorials; letters to the editor; preprints and other non-peer-reviewed publications; non-English publications; and studies published before 1 January 2018. Meta-analyses are identified during screening and retained for background, but are not included in the quantitative synthesis.

### 7.6 Context

Studies conducted in any healthcare setting involving adult thoracic transplantation, including transplant centres, hospital-based clinical practice, national or international transplant registries, and donor evaluation programmes. Studies from all countries and healthcare systems are eligible.

### 7.7 Publication date and language

Publications from 1 January 2018 onwards and published in English are eligible. No language restriction will be applied at the search stage; publication-date limits may be applied to update searches.

## 8. Information sources

**Bibliographic databases (six).** PubMed/MEDLINE (National Library of Medicine, 1966–present); Embase (Embase.com, 1947–present); Web of Science Core Collection, Science Citation Index Expanded (Clarivate Analytics); Scopus (Elsevier, 1788–present); Inspec (via Engineering Village, 1896–present); Cochrane CENTRAL (Cochrane Library Online).

**Trial registries (two).** ClinicalTrials.gov (US National Library of Medicine); WHO International Clinical Trials Registry Platform (ICTRP).

**Other methods of identifying studies.** Searching of trial and study registers. Only published studies are sought; no grey-literature search is performed and authors are not contacted for unpublished data.

## 9. Search strategy

The literature search will comprise an initial search followed by planned update searches. The same predefined search strategy will be used throughout, including identical search concepts, controlled vocabulary terms (including MeSH where applicable), and free-text search terms. Only database-specific syntax and publication-date limits will be modified for update searches. No new search concepts or eligibility criteria will be introduced in subsequent searches.

### 9.1 Concept structure

Three concept blocks are combined with `AND`:

1. **Population** — heart transplant*, heart–lung transplant*, lung transplant*, cardiac transplant*, cardiothoracic transplant*, allograft and hemograft variants, with controlled

vocabulary terms for heart transplantation, lung transplantation and heart–lung transplantation where the database provides them;

2. **Intervention** — artificial intelligence, machine learning, deep learning, neural network, convolutional neural network, random forest, decision tree, gradient boosting, support vector machine, and related terms;
3. **Outcome** — survival, mortality, rejection including acute cellular rejection, allocation, comorbidity, hospital stay, monitoring, surveillance, and modelling terms.

## 9.2 Indexing lag between searches

Because update searches may use publication-date limits, recently indexed records with earlier publication dates could be missed. To minimise this risk, update searches will include an overlap with the preceding search period, and potentially eligible records identified during this overlap will be screened according to the predefined eligibility criteria.

## 9.3 Search documentation

The searches will be documented for each information source, including the database or registry searched, the platform, the date of the search, and any publication-date limits applied. Search results will be imported into the review management system (Covidence) and deduplicated before screening. Update searches will use the same predefined search concepts and terms as the initial search, with only database-specific syntax and date limits modified as required. Complete line-by-line search strategies for all databases and registries will be provided in Appendix A.

## 9.4 Full search strategies

The complete line-by-line search strategies for all databases and trial registries are provided in Appendix A. Update searches use the same predefined search concepts and terms, with publication-date limits applied as appropriate.

# 10. Study selection

Records will be deduplicated before screening. Title/abstract screening and full-text assessment will be performed independently by at least two assessors using predefined eligibility criteria. Assessors may include human reviewers and, where used, a large language model as an independent screening tool. Discordant decisions will be reviewed by a human reviewer, whose decision will be final. Studies authored by a member of the review team will be screened, extracted, and appraised only by reviewers who are not authors of that study. Full-text exclusion reasons will be recorded systematically, and study flow will be reported using a PRISMA 2020 flow diagram.

# 11. Data collection process

Data will be extracted using a purpose-built web-based application developed by the study team, with structured fields based on a predefined extraction framework aligned with the CHARMS checklist for prediction-model reviews. Extraction will be performed independently by at least

two assessors, including human reviewers and, where used, a large language model as an independent second assessor. Discrepancies will be resolved against the source publication by a human reviewer. Key fields, including outcome definition, validation method, discrimination metric, and risk-of-bias assessment, will be verified against the original report. Study authors will not be contacted for missing information; unavailable data will be recorded as not reported. The final dataset will be locked before the definitive analysis is performed.

# 12. Data items

| Domain | Items |
|---|---|
| Identification | Study identifier; first author; year; journal; DOI; study design |
| Population | Candidates, recipients or donors; organ focus; cohort size; age; sex; enrolment period; country; data source (registry, single-centre, multicentre). |
| | Age was extracted as the study-reported mean or median. For descriptive summaries across studies, reported medians were treated as approximate measures of central tendency comparable to means. No transformation of medians to means was performed, and age was not used in the primary quantitative synthesis. |
| Model | Model family (neural network, tree-based, support vector machine, other); whether ML is the primary analytical component; number of input features; input modality (tabular, imaging, other) |
| Donor–recipient | Whether donor characteristics, recipient characteristics, or both are model inputs |
| Validation | None; internal split; k-fold; other internal; external geographic; external temporal |
| Performance | Discrimination (AUROC or fixed-time C-statistic) with 95% CI; the metric's own type; sensitivity and specificity at the reported threshold; PPV and NPV with prevalence; Brier score; accuracy with class imbalance noted; calibration where reported |
| Comparator | Whether a linear comparator model is reported, and its performance |
| Outcome | Outcome type; time horizon; effect measure and its variance |

| Domain | Items |
|---|---|
| Appraisal | Risk-of-bias rating per domain; applicability judgement; reporting-quality assessment; assessors |
| Clinical maturity | Whether the model was prospectively evaluated, externally validated, or implemented in practice |

# 13. Outcomes

## 13.1 Primary outcome

The primary outcome is predictive performance measured as discrimination, expressed as the area under the receiver operating characteristic curve (AUROC) or an equivalent fixed-time C-statistic. A concordance statistic calculated for a binary outcome at a fixed time horizon will be treated as equivalent to AUROC. A global survival concordance index measured over the entire follow-up period estimates a different quantity and will not be pooled with fixed-time AUROC estimates; such studies will be synthesised narratively. Quantitative synthesis will be performed within clinically comparable organ–outcome subgroups where sufficient data are available.

## 13.2 Additional outcomes

Additional clinical outcomes include waitlist mortality; post-transplant complications, including primary graft dysfunction and graft rejection; chronic lung allograft dysfunction, including bronchiolitis obliterans syndrome; quality of life; and health-economic outcomes. Additional model-related measures include calibration, extent and type of external validation, risk of bias, reporting quality, and clinical implementation. Complication categories will be synthesised separately and will not be pooled across different clinical outcomes. Quality-of-life outcomes will not be pooled across different instruments unless appropriate standardisation is possible. Health-economic outcomes will be synthesised narratively. Where quantitative synthesis is not appropriate or feasible, findings will be summarised narratively.

# 14. Risk of bias assessment

Risk of bias and applicability will be assessed using an adapted PROBAST-based rubric. The adapted rubric retains the four PROBAST domains (participants, predictors, outcome, and analysis) and the separate applicability assessment but applies predefined domain-level criteria rather than the original signalling questions. The analysis domain additionally requires reported validation and calibration for a low-risk rating. Reporting quality will be assessed separately using TRIPOD+AI.

Risk-of-bias assessment will be performed independently by at least two assessors, including human reviewers and, where used, a large language model as an independent assessor.

Discrepancies will be resolved by discussion against the source publication, with the final judgement made by a human reviewer.

Overall risk of bias will be classified as **low**, **unclear**, or **high**, following predefined decision rules based on the domain-level assessments.

# 15. Reporting bias assessment

Risk of bias arising from missing or selectively reported results will be considered as part of the adapted PROBAST-based assessment and the TRIPOD+AI reporting evaluation. Studies lacking the information required for quantitative synthesis may be included in the qualitative review but excluded from meta-analysis. Formal assessment of small-study effects or publication bias will not be undertaken because the expected number of studies within individual meta-analyses is limited and between-study heterogeneity is anticipated to be substantial. This domain will be considered in the overall certainty-of-evidence assessment.

# 16. Certainty of evidence

Certainty of evidence will be assessed using GRADE adapted for prediction-model performance, following published guidance for prognostic evidence. Certainty will be rated separately for each meta-analysed outcome, considering risk of bias, inconsistency, indirectness, imprecision, and publication bias. Risk of bias will be informed by the adapted PROBAST-based assessment; inconsistency by between-study heterogeneity; indirectness by differences between the included evidence and the review question; and imprecision by the uncertainty and amount of evidence contributing to each pooled estimate.

# 17. Data synthesis

All included studies will be summarised descriptively. Quantitative synthesis will be undertaken where at least three clinically comparable studies report the same organ-specific outcome with sufficient variance information. Reported 95% confidence intervals will be used where available; otherwise, standard errors may be derived from case counts using the Hanley–McNeil method. Studies lacking usable variance will be synthesised narratively. Where several eligible estimates are reported, a single estimate per study and outcome will be selected according to predefined rules, prioritising the relevant time horizon and external validation where available. For survival outcomes, where a study reports discrimination at multiple prediction horizons, the estimate corresponding to the study's explicitly defined primary prediction horizon will be selected. If no primary prediction horizon is specified, the 1-year estimate will be preferred where available. If 1-year performance is not reported, the estimate at the closest clinically comparable reported horizon will be retained for narrative synthesis or a separate time-horizon-specific analysis, as appropriate. Estimates from different prediction horizons will not be pooled solely to increase the number of studies. Potential overlap between registry cohorts will be addressed in sensitivity analyses.

# 18. Meta-analysis methods

Discrimination will be pooled using random-effects meta-analysis. AUROC estimates will be logit-transformed before pooling and back-transformed for presentation. Between-study variance will be estimated using restricted maximum likelihood (REML), with the Knapp–Hartung adjustment applied to pooled confidence intervals. Heterogeneity will be assessed using $I^2$, $\tau^2$ and Cochran's Q, and 95% prediction intervals will be reported where appropriate. Heterogeneity will be interpreted alongside the clinical and methodological differences between studies rather than by applying a fixed $I^2$ threshold.

# 19. Subgroup analyses

Primary subgrouping will be by organ and target outcome. Additional subgroup analyses will consider model family, validation type and study design. Where sufficient studies are available, these factors may be explored using meta-regression rather than separate pooled analyses. Subgroups with fewer than three studies will not be meta-analysed and will be summarised narratively.

# 20. Sensitivity analyses

Sensitivity analyses will assess the robustness of pooled estimates by leave-one-out analysis, restricting to one estimate per registry source, restricting to fixed-time binary AUROC estimates, and excluding mixed adult–paediatric populations where relevant. Additional sensitivity analyses may examine alternative within-study estimate selection where multiple eligible estimates are reported.

# 21. Meta-regression

Exploratory meta-regression will be undertaken only where at least eight studies are available within a pooled subgroup. Candidate moderators will include validation type, cohort size, number of input features, publication year, and input modality. Analyses will be interpreted as hypothesis-generating between-study associations rather than causal effects.

# 22. Missing data

Where discrimination is reported without a variance estimate, a standard error will be derived from case counts using the Hanley–McNeil method where possible; otherwise, the study will be excluded from quantitative pooling but retained in the narrative synthesis. Missing study characteristics will be recorded as not reported. No imputation of missing variance will be performed, and studies will not be excluded from the review solely because of missing data.

# 23. Software

Statistical analyses and data processing will be performed using R and/or Python. Screening and data extraction will be managed using review software, including a purpose-built web application developed by the study team. Exact software and package versions will be documented for the final analysis.

# 24. Reporting guidelines

| Guideline | Use |
|---|---|
| PRISMA 2020 | Reporting of the review and the study-flow diagram |
| PRISMA-P | Reporting and structure of this protocol |
| CHARMS | Framework for data extraction from prediction-model studies |
| PROBAST (adapted) | Risk-of-bias and applicability appraisal |
| TRIPOD+AI | Assessment of the included studies' reporting quality |
| GRADE (adapted for prognostic evidence) | Certainty of evidence |

# 25. Dissemination

Results will be submitted for publication in a peer-reviewed journal and presented at scientific meetings. The protocol, search strategies, extraction framework, risk-of-bias rubric and analysis code will be made publicly available in an appropriate repository.

# 26. Ethics

No ethical approval is required. The review uses published aggregate data, involves no individual participant data and no new patient contact.

# 27. Conflicts of interest

No conflicts of interest have been declared by the review team. Studies authored by members of the review team will be screened, extracted and appraised only by reviewers who are not authors of the respective study.

# 28. Peer review

This planned review has not undergone external peer review.

# 29. Keywords and index terms

**Keywords.** Prognostic models; prediction models; clinical decision support; risk prediction; external validation; predictive performance; clinical implementation; systematic review; meta-analysis; methodological quality.

**Medical Subject Headings.** Adult; Artificial Intelligence; Classification; Computer Simulation; Deep Learning; Graft Rejection; Heart Transplantation; Heart-Lung Transplantation; Humans; Lung Transplantation; Prognosis; Quality of Life; Random Forest; Support Vector Machine.

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# Appendix A. Full search strategies

The complete database- and registry-specific search strategies are reproduced below.

## Search strings

The same predefined search concepts and terms were used throughout; database-specific syntax was adapted where required.

### PubMed (National Library of Medicine, 1966–present)

#1 heart transplantation[Mesh] OR heart-lung transplantation[Mesh]
OR lung transplantation[Mesh] OR "heart transplant*"[tw]
OR "heart lung transplant*"[tw] OR cardiac transplant*[tw]
OR cardiothoracic transplant*[tw] OR lung transplant*[tw]
OR cardiac allograft*[tw] OR heart allograft*[tw] OR lung allograft*[tw]
OR cardiac heterograft*[tw] OR heart heterograft*[tw] OR lung heterograft*[tw]
OR cardiac hemograft*[tw] OR heart hemograft*[tw] OR lung hemograft*[tw]

#2 artificial intelligence[Mesh] OR "Machine Learning"[Mesh]
OR "Neural Networks, Computer"[Mesh] OR machine learning[tw] OR ML model*[tw]
OR neural network*[tw] OR deep learning[tw] OR decision tree*[tw]
OR "wearable sensor"[tiab:~3] OR "wearable sensors"[tiab:~3]
OR "artificial intelligence" OR AI[tw] OR random forest*[tw]
OR random survival forest*[tw] OR natural language processing[tw] OR NLP[tw]
OR chatgpt[tw] OR chat gpt[tw] OR large language model*[tw]
OR convolutional network*[tw] OR digital twin*[tw] OR gradient boost*[tw]
OR bayesian network*[tw] OR vector machine*[tw] OR naive Bayes[tw]
OR wearable device*[tw] OR applewatch[tw] OR apple watch[tw]
OR galaxy watch[tw] OR tree of predict*[tw]

#3 "Mortality"[Mesh] OR "Prognosis"[Mesh] OR "Graft Rejection"[Mesh]
OR "Immunosuppressive Agents"[Mesh] OR "Diagnostic Imaging"[Mesh]
OR "Biopsy"[Mesh] OR "Genomics"[Mesh] OR "Precision Medicine"[Mesh]
OR "Renal Insufficiency"[Mesh] OR infections[Mesh]
OR outcome*[tiab] OR donor evaluation*[tiab] OR candidate evaluation*[tiab]

OR survival[tiab] OR mortality[tiab] OR death*[tiab] OR complication*[tiab]
OR rejection*[tiab] OR prognostic[tiab] OR prognosis[tiab] OR predict*[tiab]
OR immunosuppressi*[tiab] OR matching[tiab] OR waitlist[tiab]
OR wait* time[tiab] OR image[tiab] OR imaging[tiab] OR biops*[tiab]
OR genomics[tiab] OR omics[tiab] OR personalized[tiab]
OR allograft vasculopathy[tiab] OR renal failure[tiab] OR graft failure*[tiab]
OR malignancy[tiab] OR infection*[tiab] OR allocation*[tiab]
OR graft rejection*[tiab] OR comorbidit*[tiab] OR cellular rejection*[tiab]
OR ACR[tiab] OR "hospital stay"[tiab] OR monitoring[tiab]
OR surveillance[tiab] OR modeling[tiab]

#4 #1 AND #2 AND #3
#5 #4 NOT (animals[mh] NOT humans[mh])

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## Embase (via Embase.com, 1947–present)

#01 'heart transplantation'/exp OR 'heart lung transplantation'/exp
OR 'lung transplantation'/exp OR 'heart transplant*':ti,ab
OR 'heart lung transplant*':ti,ab OR 'cardiac transplant*':ti,ab
OR 'cardiothoracic transplant*':ti,ab OR 'lung transplant*':ti,ab
OR 'cardiac allograft*':ti,ab OR 'heart allograft*':ti,ab
OR 'lung allograft*':ti,ab OR 'cardiac heterograft*':ti,ab
OR 'heart heterograft*':ti,ab OR 'lung heterograft*':ti,ab
OR 'cardiac hemograft*':ti,ab OR 'heart hemograft*':ti,ab
OR 'lung hemograft*':ti,ab
#02 'artificial intelligence'/exp OR 'machine learning'/exp
OR 'artificial neural network'/exp
#03 (free text AI/ML terms:ti,ab)
#04 #02 OR #03
#05 (outcome MeSH terms)
#06 (outcome free text:ti,ab)
#07 #05 OR #06
#08 #01 AND #04 AND #07
#09–#11 Animal filters
#12 #08 NOT animal-only

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## Web of Science Core Collection

Subsets: WOS.SCI, WOS.AHCI, WOS.ESCI, WOS.ISTP, WOS.SSCI, WOS.ISSHP

#1 TS=("heart transplant*" OR "heart lung transplant*" OR "lung transplant*"
OR "cardiac transplant" OR "cardiothoracic transplant*" OR "cardiac allograft*"
OR "heart allograft*" OR "lung allograft*" OR "cardiac heterograft*"
OR "heart heterograft*" OR "lung heterograft*" OR "cardiac hemograft*"
OR "heart hemograft*" OR "lung hemograft*")

#2 TS=("artificial intelligence" OR "machine learning" OR "neural network*"
OR "ML model*" OR "deep learning" OR "decision tree*" OR AI
OR "random forest*" OR "random survival forest*"
OR "natural language processing" OR NLP OR chatgpt OR "chat gpt"
OR "large language model*" OR "convolutional network*" OR "digital twin*"
OR "gradient boost*" OR "bayesian network*" OR "vector machine*"
OR "naive Bayes" OR "wearable device*" OR applewatch OR "apple watch"
OR "galaxy watch" OR "tree of predict*")
OR TS=(wearable NEAR/3 sensor*)

#3 (outcome terms in TS/AB/TI)

#4 #1 AND #2 AND #3

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## Scopus

#1 TITLE-ABS-KEY("heart transplant*" OR "heart lung transplant*"
OR "lung transplant*" OR "cardiac transplant*" OR "cardiothoracic transplant*"
OR "cardiac allograft*" OR "heart allograft*" OR "lung allograft*"
OR "cardiac heterograft*" OR "heart heterograft*" OR "lung heterograft*"
OR "cardiac hemograft*" OR "heart hemograft*" OR "lung hemograft*")

#2 TITLE-ABS-KEY("artificial intelligence" OR "machine learning"
OR "neural network*" OR "ML model*" OR "deep learning" OR "decision tree*"
OR ai OR "random forest*" OR "random survival forest*"
OR "natural language processing" OR nlp OR chatgpt OR "chat gpt"
OR "large language model*" OR "convolutional network*" OR "digital twin*"
OR "gradient boost*" OR "bayesian network*" OR "vector machine*"
OR "naive Bayes" OR "wearable device*" OR applewatch OR "apple watch"
OR "galaxy watch" OR "tree of predict*")
OR TITLE-ABS-KEY(wearable W/3 sensor*)

#3 TITLE-ABS-KEY(outcome terms)

#4 #1 AND #2 AND #3

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## Inspec

#1 (transplant terms WN KY)
#2 (AI/ML terms WN KY)
#3 (outcome terms WN KY)
#4 #1 AND #2 AND #3

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## Cochrane CENTRAL

#1 MeSH: Heart Transplantation (explode)
#2 MeSH: Heart-Lung Transplantation (explode)
#3 MeSH: Lung Transplantation (explode)
#4 Free text transplant terms:ti,ab,kw
#5 #1 OR #2 OR #3 OR #4
#6 MeSH: Artificial Intelligence (explode)
#7 MeSH: Machine Learning (explode)
#8 MeSH: Neural Networks, Computer (explode)
#9 Free text AI/ML terms:ti,ab,kw
#10 #6 OR #7 OR #8 OR #9
#11–#22 MeSH + free text outcome terms
#23 #5 AND #10 AND #22

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## ClinicalTrials.gov

Intervention: "heart transplantation" OR "heart transplantations"
OR "cardiac transplantations" OR "lung transplantation"
OR "lung transplantations" OR "heart transplant"
OR "cardiac transplant" OR "lung transplant"
Other terms: "artificial intelligence" OR "machine learning"
OR "neural networks" OR "neural network" OR "deep learning"
OR "wearable sensor" OR "wearable device"
OR "random forest" OR "decision forest"

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## ICTRP / WHO

Title: artificial intelligence OR machine learning OR neural network
OR neural networks OR deep learning OR wearable sensor
OR wearable device OR random forest OR decision forest
Intervention: heart transplantation OR heart transplantations
OR cardiac transplantations OR lung transplantation
OR lung transplantations OR heart transplant
OR cardiac transplant OR lung transplant

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